A-A-51160A
December 12, 1991
SUPERSEDING
A-A-51160
October 16, 1986

COMMERCIAL ITEM DESCRIPTION

FOLDING MACHINES, SHEET METAL, BAR TYPE, HAND OPERATED, BENCH MOUNTED

The General Services Administration has authorized the use of this commercial item description.

1. ABSTRACT

1.2 Abstract. This description covers bench mounted, hand operated, bar type, sheet metal, folding machines of the standard, heavy duty, commercial type which are used in sheet metal fabricating and repair shops.

2. CLASSIFICATION

2.1 Classification. The folding machines shall be of the following sizes. The size to be furnished shall be as specified (see 7.1 and table I).

Size 1 - 22 gauge x 30 inches long.
Size 2 - 20 gauge x 20 inches long.
Size 3 - 20 gauge x 30 inches long.
Size 4 - 20 gauge x 36 inches long.
Size 5 - 20 gauge x 42 inches long.

3. SALIENT CHARACTERISTICS

3.1 Design. The folding machines shall be new and of the manufacturer's commercial design in effect on the date of the solicitation. The machines shall have provisions for bench mounting.

3.2 Frame. The frame shall have the strength and rigidity to allow the machine to produce bends of a consistent size and form.

Beneficial comments recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd., Memphis TN 38114-5213.

AMSC N/A FSC 3441

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TABLE I. Folding machine capacities.

Machine size	Capacities		
	Gauge number	Folding length (inchs)	Fold width gauge range (inch)
1	22	30	1/8 to 1
2	20	20	1/8 to 1
3	20	30	1/8 to 1
4	20	36	1/8 to 1
5	20	42	1/8 to 1

NOTE: Where a range is shown, the value shall be not greater than the lower and not less than the greater value.

3.3 Folding blade. The folding blade shall be of sufficient length to comply with the requirements of table I for the machine size specified. The machines shall be provided with a counterbalance arrangement to balance the folding bar.

3.4 Clamping jaw. The clamping jaw shall be actuated by the operating lever and shall grip and hold the sheet metal for all bending operations.

3.5 Stops. The folding machines shall be provided with two positive stops, one to form 45 degree bends and the other to form 90 degree bends, and one adjustable stop to permit forming any desired angle.

3.6 Fold width gauge. The folding machines shall be provided with an adjustable gauge for controlling the width of the folds. The gauge shall be graduated in increments of 1/16 inch over the range specified in table I. A thumb screw or other means shall be provided for locking the gauge at the desired setting.

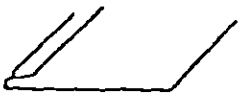
3.7 Wing. The wing shall be of the size and strength to perform all operations within the capacity of the machine. The wing shall be of the adjustable type with the adjustment provided by a wedge or other suitable means. Wing adjustment shall be capable of being set for forming sharp bends and locks and can be adjusted for open locks or rounded bends.

3.8 Operating lever. The operating lever shall be of the length and strength to accomplish the task of the size machine specified.

3.9 Performance. The folding machines shall be capable of making bends or folds to the edges of sheet metal of the widths and thicknesses, specified in table I. The bends or folds shall be similar to those found in figure 1 through figure 7. The machine shall be capable of forming mild steel with a tensile strength of 60,000 pounds per square inch.

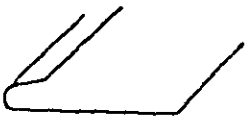
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Figure 1.



SHARP LOCK

Figure 2.



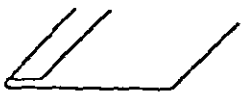
OPEN LOCK FOR WIRE EDGE

Figure 3.



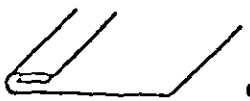
90 DEGREE FLANGE

Figure 4.



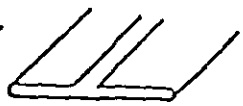
CLOSED LOCK OR SINGLE HEM

Figure 5.



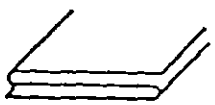
DOUBLE HEM

Figure 6.



DRIVE CLEAT

Figure 7.



"S" CLIP

3.10 General requirements.

3.10.1 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request shall be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

3.10.2 Environmental protection. During the manufacture of the folding machine, materials hazardous to the ecological system as prohibited by the Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

3.10.3 Reference documents. The documents referenced in this document shall be the issue in effect on the date of issuance of the invitation for bid or request for proposal.

4. REGULATORY REQUIREMENTS

4.1 Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

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5. QUALITY ASSURANCE

5.1 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract .

6. PRESERVATION, PACKAGING, LABELING, AND MARKING

6.1 Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this commercial item description.

b. Size and quantity required.

7.2 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

A51160-X

- X: Size:
  - 1 - 22 gauge x 30 inches long
  - 2 - 20 gauge x 20 inches long
  - 3 - 20 gauge x 30 inches long
  - 4 - 20 gauge x 36 inches long
  - 5 - 20 gauge x 42 inches long
- A51160: CID number

7.3 Cross reference of classification changes. Classification changes from the previous issue are as follows:

A-A-51160	A-A-51160A
Size 0	(Deleted)
Size 1	Size 1
Size 2	Size 2
Size 3	Size 3
Size 4	Size 4
Size 5	Size 5

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7.4 Source for documents referenced in this CID.

Federal standard, FED-STD-376 is available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

MILITARY INTEREST:

Custodian

Air Force - 99

Review Activities

Air Force - 84
DLA - GS

User Activity

Navy - SH

CIVIL AGENCY COORDINATION ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

DLA - IP

Project 3441-0153

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